Software Requirements Specification

Hiking Tour Assistant: software for a dual component system of Raspberry Pi and LilyGoT-Watch 2020 V2

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1. Introduction

1.1. Purpose

The purpose of this Software Requirements Specification (SRS) is to provide a comprehensive description of the functional and non-functional requirements for the Hiking Tour assistant. This document establishes the technical foundation for software of a dual-component system consisting of a wearable smartwatch application for real-time hiking data acquisition and a Raspberry Pi-hosted web interface for data visualization.

This document is intended for software developers and the ELEC-E8408 Embedded Systems Development course staff.

1.2. Scope

This specification covers the development of the Hiking Tour Assistant software for the dual-component system, which comprises a LilyGoT-Watch 2020 V2 smartwatch and Raspberry Pi single-board computer.

The smartwatch will record real-time hiking data, including step count, traveled distance and session duration. After a session has ended, the data will be sent using bluetooth to the Raspberry Pi. The Raspberry Pi provides a browser-based local interface for a user to review the latest hiking session data.

This system is intended for personal data logging. The primary purpose is to provide a hiker a self-hosted, easily modifiable alternative to commercial devices.
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1.3. Definitions, acronyms, and abbreviations

Acronyms & Abbreviations

Smartwatch	LILYGO T-WATCH-2020 V2
External device	Rasberry Pi single-board computer
SRS	Software Requirement Spesification
WebUI	Web User Interface

Definitions

Raspberry Pi: A small, single-board computer used in this project as the central data storage and web host.
Session: A single, continuous recording event initiated by the user on the smartwatch. A session begins when there is no active session and the button is pressed and it terminates upon the next button press.

1.4. References

The following documents provide supporting information for the requirements defined in this SRS. In the event of a conflict between this document and the referenced materials, this SRS shall take precedence.

Title	Date	Publishing Organization	Source
IEEE Std 830-1998	Oct 1998	IEEE Computer Society	https://ieeexplore.ieee.org/document/720574

1.5. Overview

This SRS is organized into three sections to guide the development and analysis of the hiking data system:

- **Section 1: Introduction** provides the purpose, scope, and basic definitions to establish a common language for the project.

- **Section 2: Overall description** describes the general factors that affect the Hiking Tour Assistant and its requirements, including the product perspective, high-level functions, user characteristics, and technical constraints.

- **Section 3: Specific requirements** contain all the software requirements at a level of detail sufficient to enable designers to design the Hiking Tour Assistant to satisfy those requirements.

- **Appendixes and Index** provide additional supporting data and a quick-reference guide for key terms and requirements locations.

2. Overall description

Hiking Tour Assistant includes an application for the smartwatch that is able to record and display hike session statistics, such as: travelled distance, step count and hiking route records.

In order to analyze results and stay motivated, a software for an external device is developed which provides a webUI that allows the user to observe collected trip statistics.

2.1. Product perspective

2.1.1. System interfaces

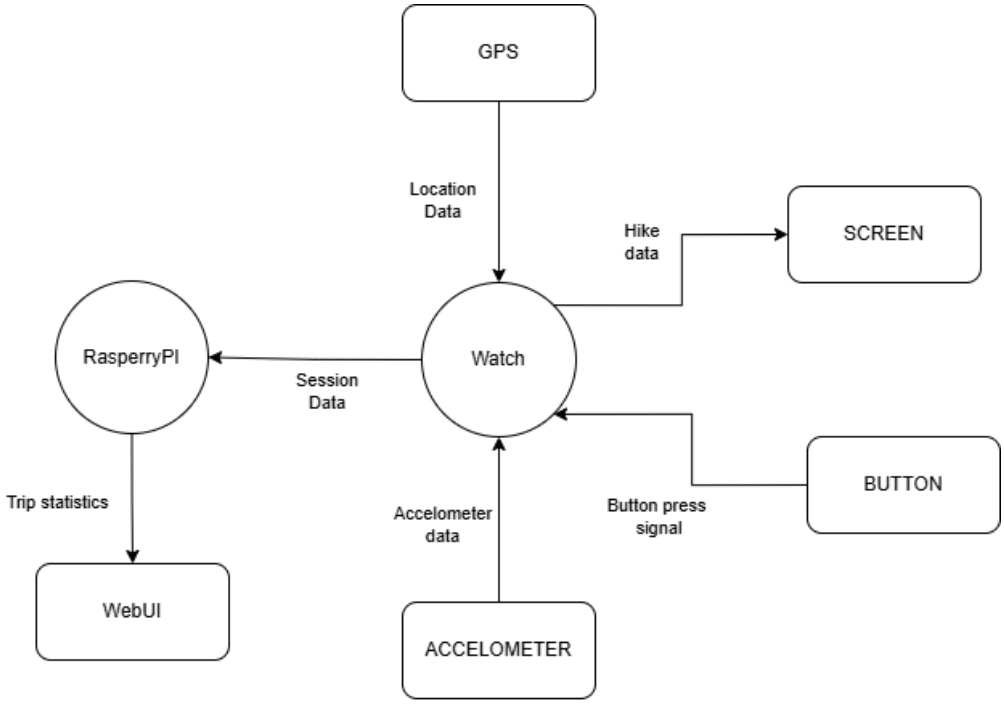

Figure 1, Context diagram of the system

2.1.2. User interfaces

smartwatch screen and side button

The user communicates with a smartwatch using the screen and the button. Button can be pressed to start and stop a session.

The smartwatch screen displays the data interface for the user.

WebUI
The User can access the webUI with a web browser. There, the statistics are displayed from the latest session. This includes the travelled distance and the step count.

2.1.3. Hardware interfaces

Wireless communication between the smartwatch and the external device
The smartwatch sends session data to the Raspberry Pi. This communication is wireless and doesn't require the user to initiate it.

Watch
Handles current hike session functionality and transferable data to Raspberry PI.

GPS
Sends GPS location data to the smartwatch.

Accelerometer
Sends accelerometer data to the smartwatch.

Button
Sends a user-input signal to smartwatch for application and smartwatch functionality. Provides shutdown and startup for smartwatch. Provides reset, start and end functionalities for the application.

Screen
Provides a user-interface to the smartwatch

Raspberry PI
Stores all transferred hike session data from smartwatch and provides Web UI-functionality.

2.1.4. Software interfaces

The Hiking Tour Assistant software on Raspberry Pi operates under Raspberry Pi OS. The Hiking Tour Assistant is compatible with Firefox web browser.

2.1.5. Communications interfaces

The Hiking Tour Assistant uses Bluetooth to communicate between the Raspberry Pi and the smartwatch.

Web communication

The Raspberry Pi hosts a server that communicates with the web browser to produce the WebUI.

2.1.6. Memory

The Raspberry Pi has 1 GB RAM and 32 GB SD card memory. The smartwatch has Flash Storage of 16MB, PSRAM (Extended RAM) of 8MB and SRAM (Internal): 520KB.

2.2. Product functions

Hiking tour assistant provides following main functions:

1. Start & stop hiking sessions
2. Record steps count and convert into travelled distance during the session
3. Display this data on a smartwatch screen
4. Synchronize and store data with Raspberry Pi via Bluetooth
5. Calculate estimated amount of calories burned during the session on Raspberry Pi
6. Initialize Web UI and show last session statistics (travelled distance, step count and burned calories)

2.3. User characteristics

The Hiking Tour Assistant is designed for various kinds of users ranging from fitness enthusiasts to tech hobbyists. Some, but not all, typical users are 1. privacy-conscious hikers, 2. tech hobbyists, 3. academics.

1. The hikers are physically active individuals who require real-time feedback and ability to review the data later from a larger screen. They value privacy and data ownership over commercial cloud-based alternatives.
2. The tech hobbyists are comfortable interacting with non-standard hardware like the LilyGo T-Watch and the Raspberry Pi. They expect the system to be modular

and modifiable. They may want to tweak the code, add new sensors, or customize the WebUI.

3. The academics include the course staff. They require the system to be stable enough for evaluation and documentation that explains how the hardware components interact.

2.4. Constraints

The software is constrained by hardware limitations described in section 3.5. smartwatch shall not perform heavy data analysis locally. Analysis shall be made by external device.

The webUI shall be accessible by the standard web browser Firefox.

2.5. Assumptions and dependencies

The Hiking Tour Assistant assumes several components and technologies to operate and depends on several technologies and components.

2.5.1 Assumptions

The Hiking Tour Assistant assumes the following components and technologies:

Hardware Availability - Smartwatch
The LILYGO T-WATCH-2020 V2 will have:

- Functional components
- Battery with 350 mAh capacity

Hardware Availability - Raspberry PI
The Raspberry Pi will:

- Have functional components
- Operate with stable power supply
- Have a SD card with 32 GB capacity

Operating System Availability
The Raspberry Pi operation system will run a Raspberry Pi OS capable of:

- Running a lightweight database
- Hosting a local web server
- Support Bluetooth communication
- Support external screen

Bluetooth Connectivity

- A stable Bluetooth connection will be available between the smartwatch and Raspberry Pi.

Sensor Accuracy

- The accelerometer and GPS sensors will provide sufficient (target values?) accurate data for estimating step count and traveled distance. Minor inaccuracies are acceptable for the system.

Single-User System

- The system is assumed to be used by a single user one at a time on a unique session.

User Environment

- The user understands basic interaction with the smartwatch and web interfaces.

2.5.2 Dependencies

The Hiking Tour Assistant depends on the following external components and technologies:

Hardware

- LILYGO T-WATCH-2020 V2 hardware components.
    - Accelerometer
    - GPS module
    - Screen (LCD)
    - Battery

  Button
    - Bluetooth module
- Raspberry Pi hardware components.
    - Bluetooth module
    - SD memory storage
    - CPU
    - RAM
    - OS

Bluetooth

- Bluetooth support on smartwatch and Raspberry Pi.

Web Browser Compatibility

- The WebUI depends on compatibility to modern versions of Firefox

Operating System

- Updates might affect compatibility with the application software

Power Supply

- The smartwatch battery condition affects maximum session duration and operating time.
- The Raspberry Pi must have uninterrupted power during data transfer to prevent corruption. The WebUI operation relies on uninterrupted power on the server host.

Storage Availability

- SD card storage, at least 32 GB must be available to store session data
- SD card corruption or failure may lead to data loss

3. Specific requirements

3.1. External interfaces

User interfaces of watch

Name of Item	Smartwatch screen	Smartwatch button
Description of purpose	Output display as visual interface	Input interface allowing control of watch
Source of input or destination of output	Output for user	User input
Valid range, accuracy, and/or tolerance	-	-
Units of measure	-	-
Timing	Refresh rate of <100ms	
Relationships to other inputs/outputs	-	-
Screen formats/organization	1.54 inch LCD	-
Window formats/organization Data formats	-	-
Command formats	-	Single press: select Long press >2s: turn device on/off
End messages	-	-

User interfaces of Raspberry Pi

Name of Item	WebUI
Description of purpose	Display session statistics
Source of input or destination of output	Output for user

Valid range, accuracy, and/or tolerance	-
Units of measure	-
Timing	-
Relationships to other inputs/outputs	-
Screen formats/organization	-
Window formats/organization Data formats	-
Command formats	-
End messages	-

Wireless communication between smartwatch and external device

Name of Item	Wireless communication between smartwatch and external device
Description of purpose	Send session data for external device
Source of input or destination of output	Source: smartwatch Destination: External device
Valid range, accuracy, and/or tolerance	range >2m
Units of measure	-
Timing	1 min requirement for successful data transmission
Relationships to other inputs/outputs	-
Screen formats/organization	-

Window formats/organization Data formats	-
Command formats	-
End messages	-

3.2. Functions

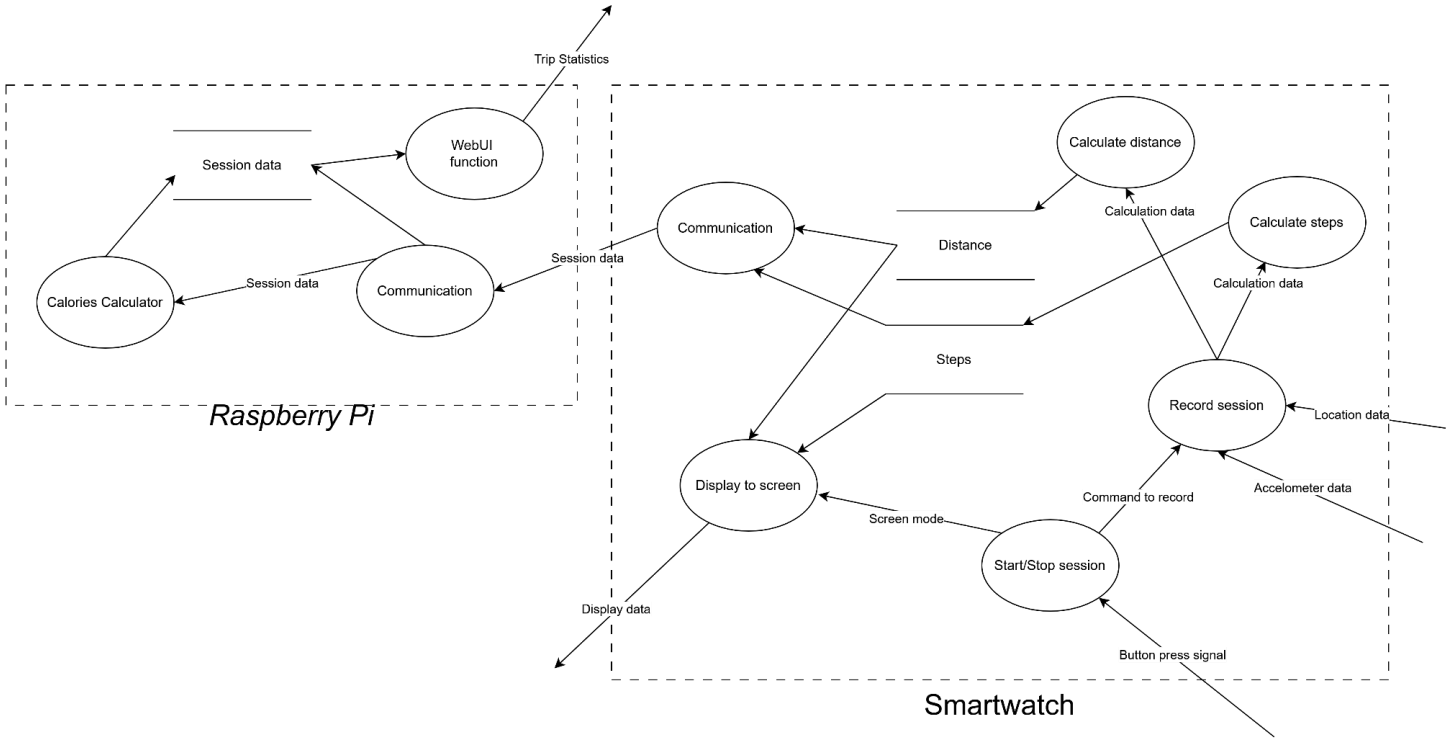

Figure 2, Data flow diagram

Smartwatch

- Display to screen
- Start & stop session
    - Single press starts
    - Single press stops
- Record session
    - Send session data for distance and step calculation
- Calculate steps

- Calculate distance
- Communication
    - Send session data for external device

Raspberry Pi

- Communication
    - Send session data for calories calculation and database
- Calculate calories
- WebUI function
    - Initialize WebUI and show session statistics

3.3. Performance requirements

- Saving a step and preparing to save the next one must take a maximum of 150 ms.
- Watch must save only 1, the latest session.
- The session can be 24 h long.
- Watch UI must react to user input in 100 ms.

3.4. Logical database requirements

Defines the logical requirements for all information stored in the system database hosted on the Raspberry Pi. The database supports storage, retrieval and processing of hiking session data transferred from the smartwatch.

The database is intended for local use only and supports the WebUI and backend processing functions.

Types of information used by various functions;

The database stores the following types of information:

- Raw session data (received from smartwatch):
    - Session start time
    - Session end time
    - Step coin
    - Calculated distance
    - Session duration

- Derived data (calculated on Raspberry Pi):
    - Estimated calories burned
- System metadata:
    - Unique session identifier
    - Timestamp of data reception
    - Data transfer status (optional, intended for error handling)

Frequency of use;

- Write operations:
    - Occur once per completed session (maximum on session per 24 hours in normal usage).
- Read operations:
    - The latest session is retrieved each time the WebUI is loaded.
    - Read frequency depends on user activity but is expected to be low to moderate (personal use application).
- Update
    - Derived values (e.g., calories burned) are written once immediately after session insertion.
    - No continuous updates are required.

Accessing capabilities;

The database supports the following access capabilities:

- Insert
    - Insert a complete session record after successful Bluetooth transfer.
    - Insertion is atomic (either all session data is sent or none is accepted).
- Read
    - Retrieve the most recent session.
    - Retrieve a certain session.
    - Retrieve all stored sessions (future implementation).
- Delete (Optional/Future)
    - Manual deletion of sessions via administrative tool.
- Access control
    - Only the backend application is directly in access with the database.
    - The WebUI access data only through backend APIs.
    - Direct database access via browser is not permitted.

Data entities and their relationships;

- Primary Entity: **Session**
    - Each session represents one completed hiking event.
- Session Entity attributes
    - session_id (Primary key)
    - start_time

    - end_time
    - duration_seconds
    - step_count
    - distance_meters
    - calories_burned
    - received_timestamp

Integrity constraints;

- Primary Key Constraint
    - session_id must be unique and auto-incremented.
- Entity integrity
    - Calories can be NULL.
    - Other values are not NULL.
- Domain constraints
    - distance_meters >= 0.
    - step count >= 0.
    - duration_seconds >= 1.
    - calories_burned >= 0 or NULL.
- Temporal Consistency
    - end_time must be later than start_time.
    - duration_seconds must be equal to equation (end_time - start_time).
- Atomicity requirement
    - Corrupted or partial session transfers do not create incomplete database entries.

Data retention requirements.

- The Raspberry PI stores multiple historical sessions on a database file until storage capacity is reached.
- The system shall support storage of at least 1000 session records within available SD card capacity.
- No automatic deletion policy is required until further implementation.
- Manual backup is possible by copying the database file.
- Data remain stored across system reboots.

The smartwatch stores only the most recent session, but the RaspberryPi maintains long-term storage for analysis and review.

3.5. Design constraints

There are no applicable standards.

Hardware limitations:

- Raspberry Pi 3 Model B+
    - 1 GB RAM
    - 32 GB SD card memory
    - 1,4 GHz Broadcom BCM2837B0 64-bit ARM Cortex-A53 Quad-core CPU
    - Communication
        - Bluetooth 4.2, Bluetooth Low Energy (BLE)
        - 2.4/5 GHz dual-band 802.11ac Wi-Fi (100 Mb/s)
        - 300 Mb/s Ethernet RJ45 with PoE support
    - micro-USB power

- LILYGO T-WATCH-2020 V2
    - Main Chip: ESP32
    - 1.54 inch LCD Capacitive touch screen: ST7789V
    - Lithium battery: 350ma
    - Triaxial accelerometer: BMA423
    - Touch Screen Chip: FT6236U
    - IR Transmitter
    - PMU Power Management: AXP202
    - GPS L76K, TF, Card Holder
    - DRV2605L Haptic Driver
    - RTC Clock Module: PCF8563

3.6. Standards compliance

This system is in compliance with no standards.

3.7. Software System Attributes

3.7.1. Security

Unauthorized access to the webUI shall be prevented. The webUI shall require a password before granting access to any personal data. Data transmission is done via Bluetooth, which is responsible for securing the connection.

3.7.2. Reliability

If the watch loses connection to the Raspberry Pi, the latest session data is shown by the Raspberry Pi until a new session is stored in the Raspberry Pi. If the Bluetooth

connection is lost in the middle of a data transmission, the data is fully resent when the connection returns.

Both the Raspberry Pi and the watch store one session's data as long as it is not overwritten by new session data.

3.7.3. Usability

All the hiking features (start, stop) are easily accessible via the physical side button. The button is usable even if the user is wearing gloves. The watch reacts to user input within 100 ms.

3.7.4. Maintainability

One subgroup of the target audience, tech hobbyists, expects a modifiable system where they can tweak the code or add new sensors, and another, the course staff, expects easy evaluability from the system. The codebase is therefore modular and well-documented.

3.7.5. Portability

The webUI must work correctly on the standard browser Firefox. Deployability to other Raspberry Pi models or watches is not guaranteed.

3.7.6. Sustainability

The watch has an easily replaceable battery in case the battery runs out during a session.

Appendix (optional)

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